

YrkesCo Database Design

- From Conceptual to Physical Model
- Docker & PostgreSQL Implementation

Business Requirements

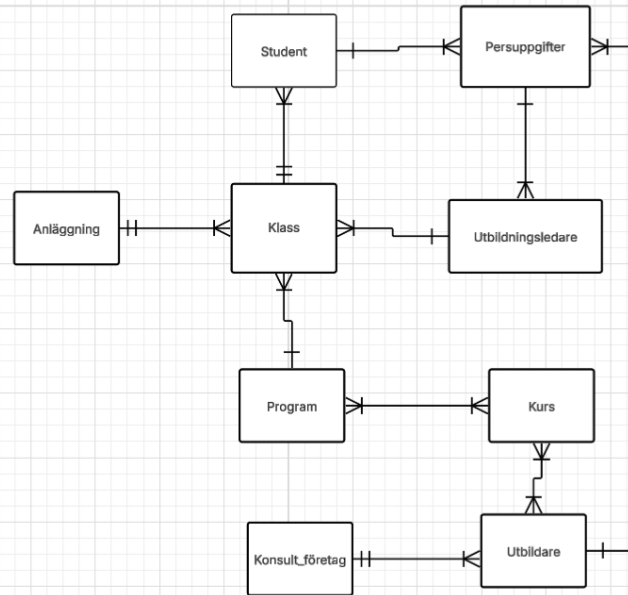
Based on the problem description, the following business requirements were identified:

- Store information about students and their personal data
- Manage programs, classes, and courses
- Track education leaders and educators
- Support educators working as external consultants
- Handle multiple locations
- Separate sensitive personal data into dedicated entities

Data Modeling Approach

- Conceptual model
- Logical model
- Physical model

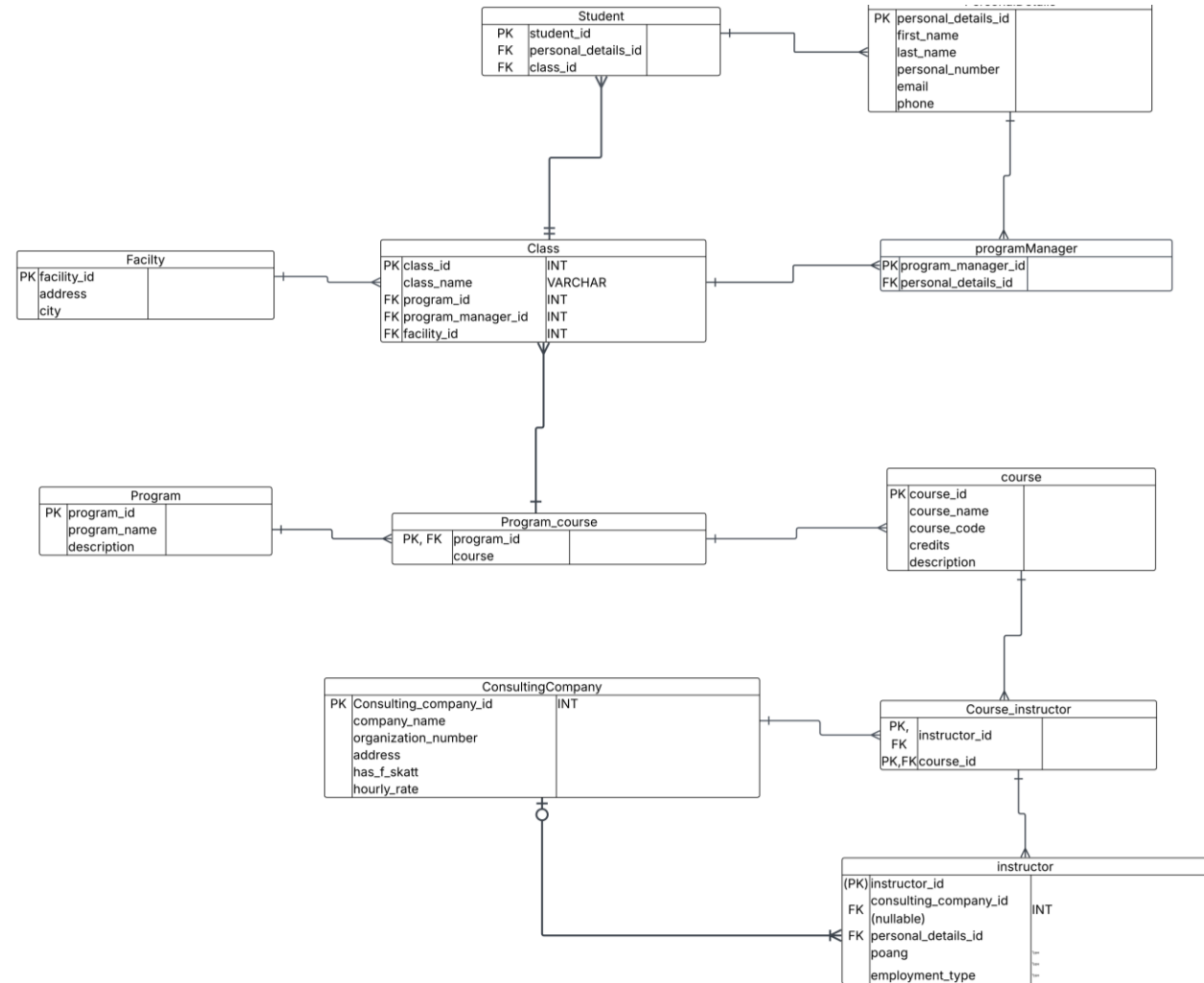
Conceptual model



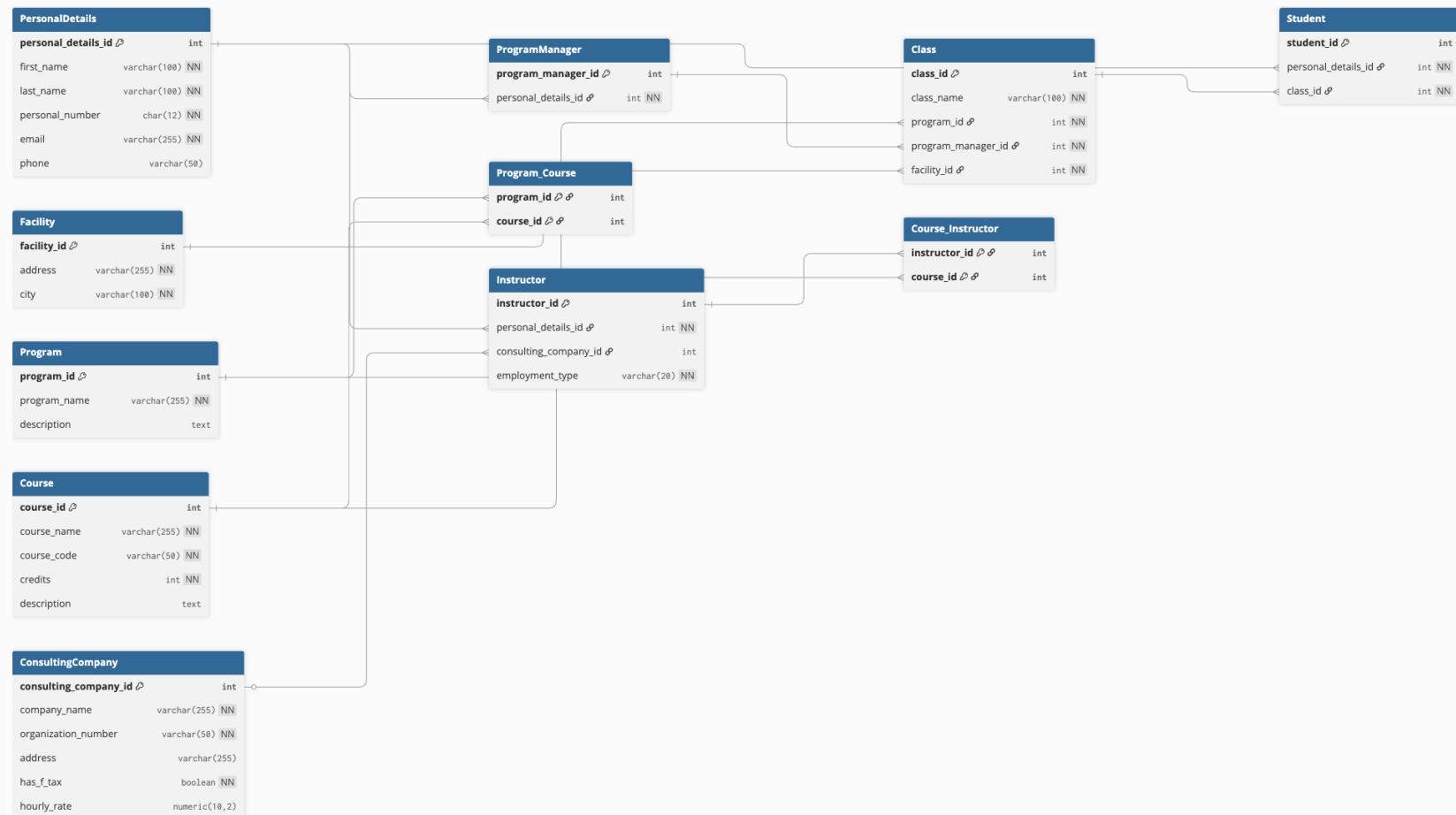
Logical model

Design Decisions

- Program_Course
- Course_Instructor
- Separate PersonalDetails
- Optional consulting company



Physical model



Logical to Physical Model

- PostgreSQL implementation
- Data types
- Constraints
- Foreign keys

Normalization & Data Integrity

- The database is normalized to Third Normal Form (3NF):
- First Normal Form (1NF): all attributes are atomic
- Second Normal Form (2NF): all non-key attributes depend on the full primary key
- Third Normal Form (3NF): no transitive dependencies

Conclusion

- This presentation demonstrates a complete data modelling process from conceptual design
- to physical implementation. By structuring the presentation from an overall business
- perspective to technical details, the resulting database solution meets YrkesCo's
- requirements while remaining scalable, normalized, and suitable for future development.